

MPLS Japan 2022
Evolution of Whitebox Solution

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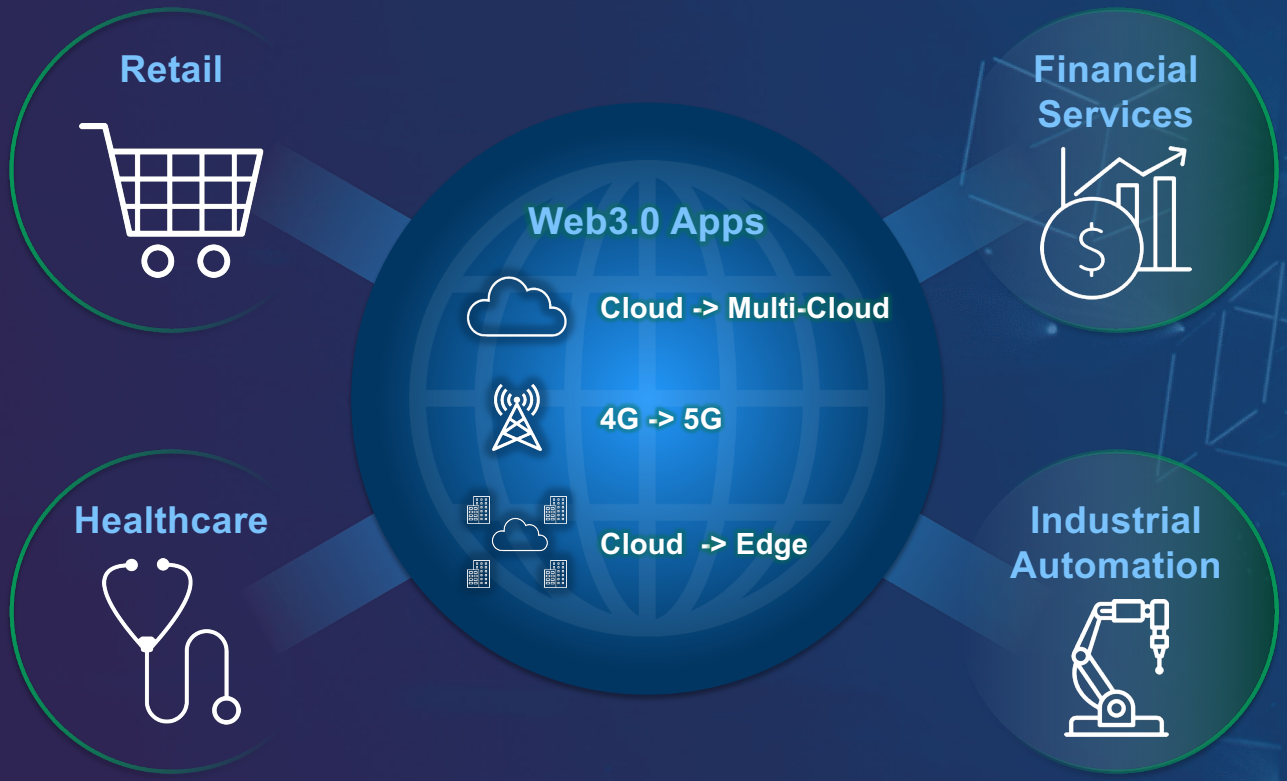
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Agenda

- **Use case**
- **Integration with Open Source Software (SONiC)**
- **Challenge on scalability, performance, security**
- **Summary**
- **Q&A**

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Enterprises Need Agility and Flexibility in A Hybrid World



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However … Legacy Networks are a Huge Bottleneck

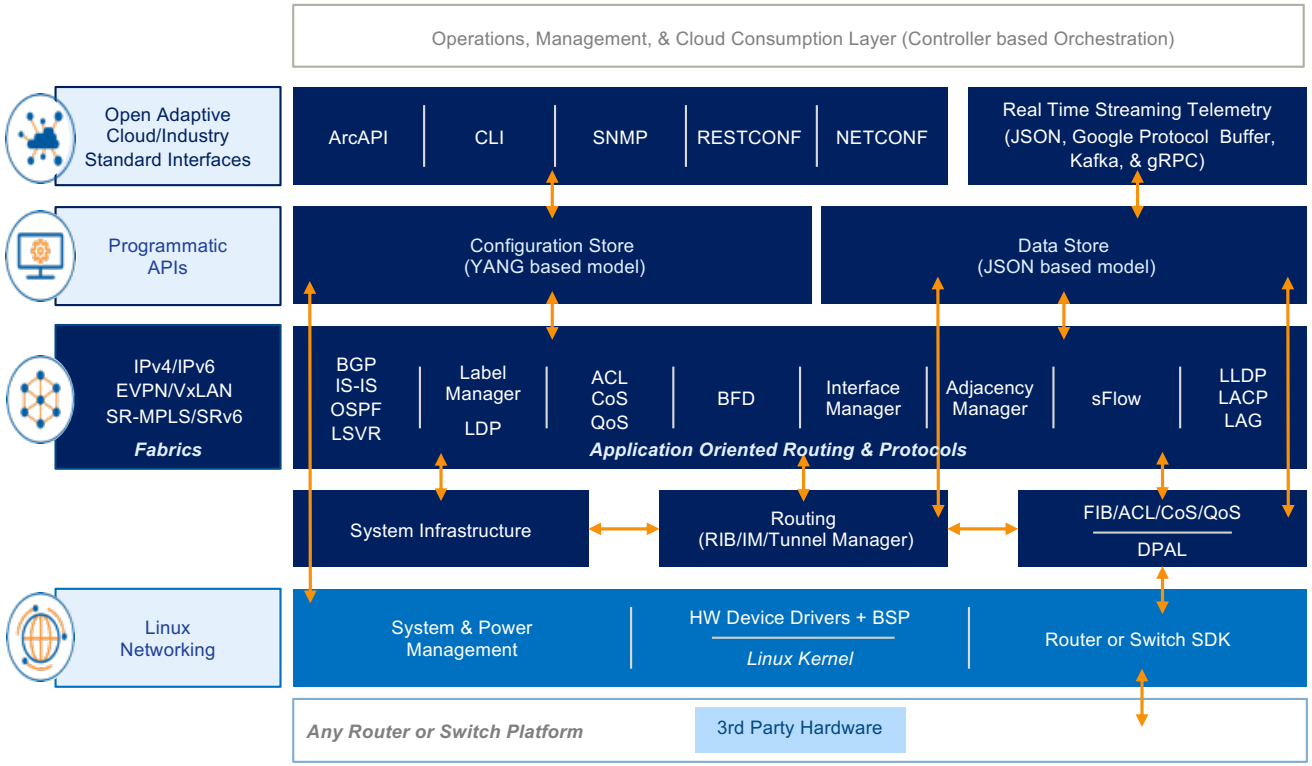


Problem with current Transport Networks

- Expensive
- Inflexible
- Not Software-defined
- Not Multi-cloud, 5G, Edge ready

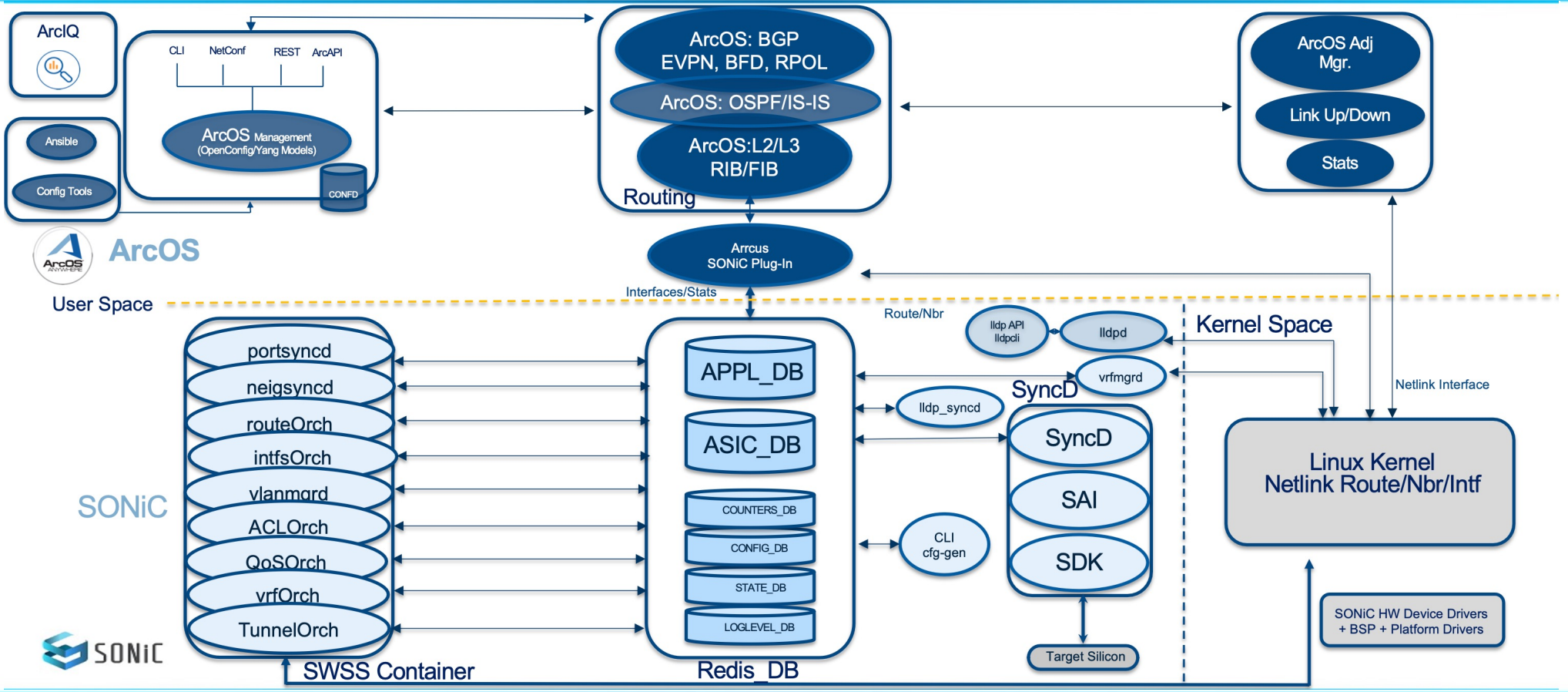
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Commercial Software Structure for Whitebox



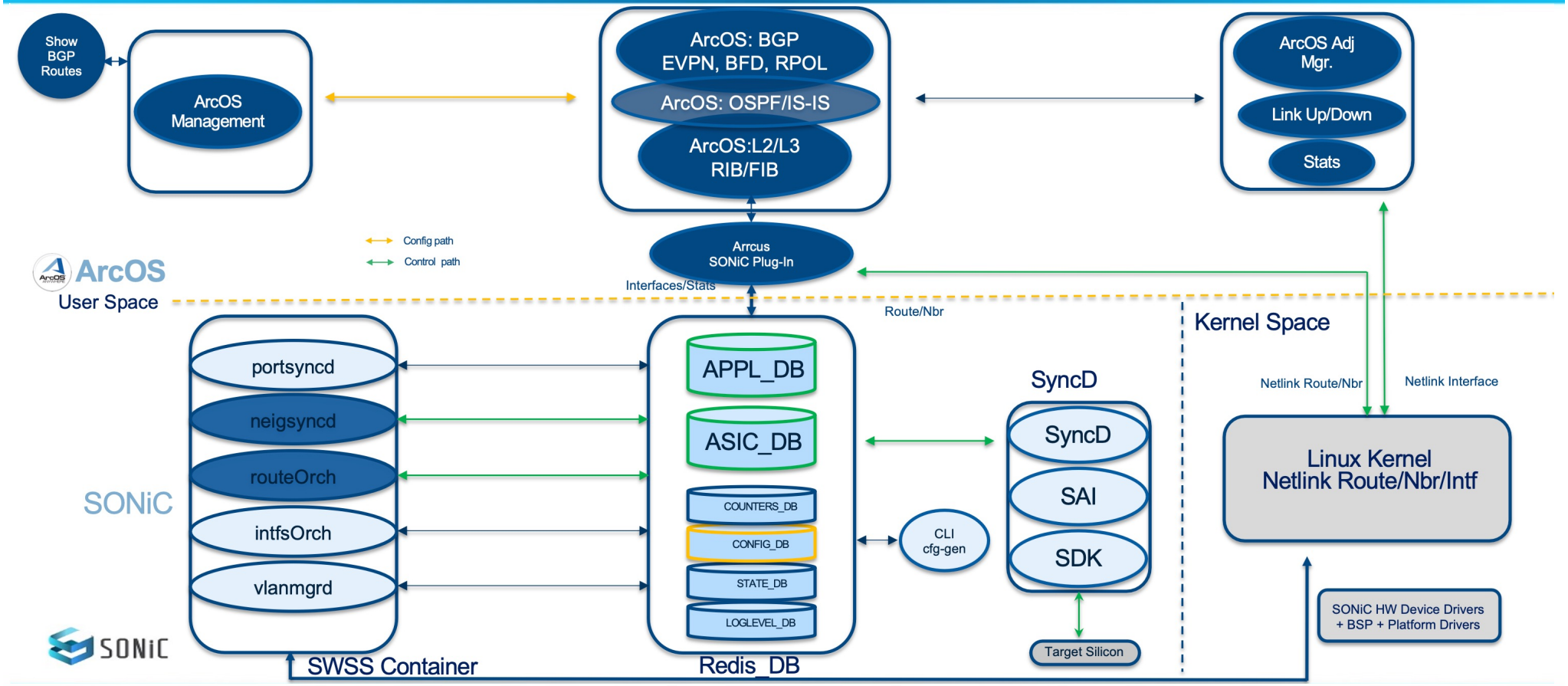
- **Microservices Architecture**
  - Components implemented in user space
  - Minimal required support of Linux Kernel
  - Container ready
- **Modularity**
  - Replace/Add/Use any component with minimal impact
  - Every component implemented as a multi-threaded separate process
- **Hardware Agnostic**
  - Router, Switch and/or Server ready
  - Intel and/or ARM processor support
  - Dataplane Adaptation Layer (DPAL)

Integration with OSS (SONiC): Architecture



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Integration with OSS (SONiC): Routing



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Challenge on Whitebox software with OSS

- **Scalability/Performance Measurement**
- **Security Issue**

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Scalability/Performance Measurement

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ArcOS/ArcOS+SONiC Convergence Measurement

Topology

IXIA -----Router#1 ---- (8 ebgp session)— 7326-56X(ArcOS w/ SONiC/ArcOS) ---------- Router#2

BGP Profile	ArcOS Convergence Time	ArcOS + Sonic Convergence Time	Comments
30K BGP IPV6 host routes learnt from 8 ebgp sessions (8 ways ecmp/ multipath)	**1.2 secs**	**2.3 secs**	BGP Convergence (Control plane)
	4 secs	**15 secs**	Forwarding Convergence (Data plane)

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ArcOS/FRR Convergence Measurement

Topology

IXIA --------- UUT (FRR/ArcOS) ---------- Router

BGP Profile	ArcOS Convergence Time	FRR Convergence Time	Comments
32 BGP Sessions/ 900K Prefixes/ 28.8M Paths Profile Used … •32 BGP Sessions on IXIA •Each BGP Session advertising same 900K prefixes	**50 secs**	**5 mins 10 secs**	Test Methodology: Route convergence measured by bringing up interfaces between IXIA and UUT. This convergence number is an average of 5 iterations.
	44 secs	**4 mins 30 secs**	Test methodology: Keep BGP sessions up and measure convergence by only advertising routes. This convergence number is an average of 5 iterations.

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Scale and Performance Analysis on XGS family

ArcOS on IX8		
Routes	With Acks	No Acks
50k v4	7.91 s ~6.3 k/sec	.678 s ~73 k/sec

Local Programming Rate with/without HW Route Ack

ArcOS on IX8		Others on TD3	
1 v4	15-25 ms	1 v4	30-50 ms

E2E Route Advertisement Latency With Route Ack

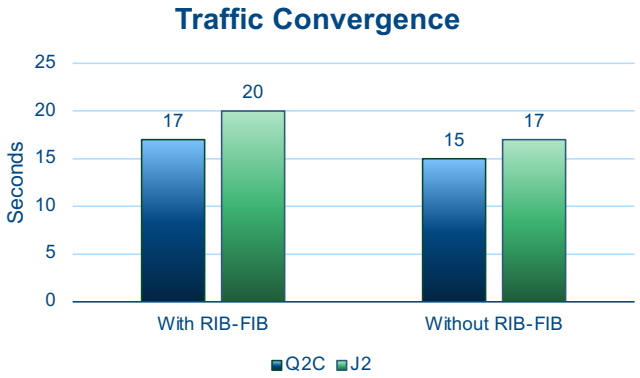
ArcOS on IX8			Others on TD3		
Routes	Acks	No Acks	Routes	Acks	No Acks
40K v4	5.8-6.2s	0.6-0.7s	40K v4	12.75-13s	2.2-2.3s
40K v6	8.5-9s	0.8-0.9s	40K v6	16.5-17s	2.2-2.25s
250K v4	39-40s	8.5-9.5s	250K v4	78-80s	15-15.5s

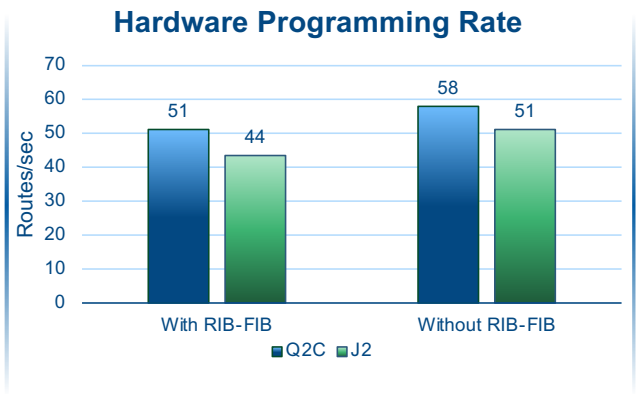
Network Convergence Time

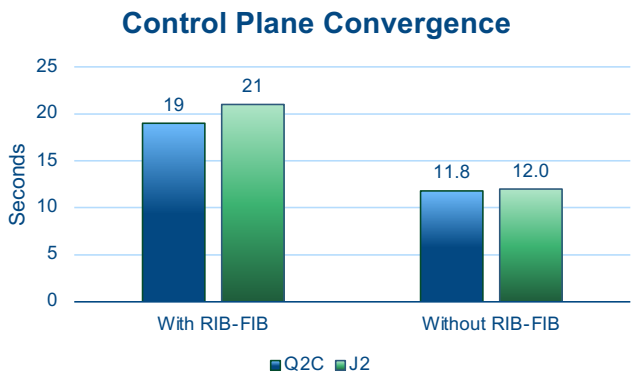
- Definition
  - *HW Route Ack* = explicit Ack from HW when a route is installed
- Goals
  1. Measure local Route Programming Rates with and without HW Route Ack enabled
  2. E2E BGP route advertisement latency with HW Route Ack enabled
     - Measure time taken to advertise a SINGLE BGP route received when it is being held till route is programmed in HW and an explicit Ack has been received
  3. Network convergence N1-DUT-N2
     - For X prefixes, measure (last update received by N2 – last update sent by N1) once BGP neighbor from DUT to N1 is unshut
- Measurements done on TD3 HW, ALPM mode, subnet routes

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Routing Performance Analysis for DNX family







Traffic Convergence

- Under 30 seconds traffic convergence
- Full Internet Route Table (IPv4 + IPv6)

Hardware Programming

- Up to 58k routes/sec
- Full Internet Route Table (IPv4 + IPv6)

Control Plane Convergence

- Under 12 seconds convergence
- Full Internet Route Table (IPv4 + IPv6)

One Architecture. One Software

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Challenge to use OSS for Whitebox

- Lacks support for Holistic CLI interface
  - Includes lack of support for programmatic APIs
- Lacks Telemetry at scale
- Less performance/scalability
- Lacks support for Hardware resource usage
- Lacks Routing support for DNX

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Scalability/Performance Analysis

- **SAI API is slower than the traditional SDK**
- **SDK might not be optimized even though using the same SDK (there are some spaces to improve the performance).**
  - However, SDK is not disclosed at all… In addition, SDK is not available for all available chipsets.

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Security Issue

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Security issue

- **Sometimes, someone running out of the device are using the local service running inside of the device like syslog, ntp, etc. As the result, a lot of resources are consumed on the device and some services are going down…**
- **Someone might steal the information from the device or might inject the invalid routing/operation to the device…**

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Why…

```
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
udp        0      0 0.0.0.0:55015           0.0.0.0:*                           1879/rsyslogd
udp        0      0 0.0.0.0:56204           0.0.0.0:*                           2946/rsyslogd
udp        0      0 0.0.0.0:123             0.0.0.0:*                           3194/ntpd
udp        0      0 0.0.0.0:49591           0.0.0.0:*                           3200/rsyslogd
udp        0      0 0.0.0.0:33435           0.0.0.0:*                           3217/rsyslogd
udp        0      0 0.0.0.0:50172           0.0.0.0:*                           5613/rsyslogd
udp        0      0 0.0.0.0:58719           0.0.0.0:*                           3290/rsyslogd
udp        0      0 0.0.0.0:44045           0.0.0.0:*                           2642/rsyslogd
udp        0      0 0.0.0.0:60505           0.0.0.0:*                           5634/rsyslogd
udp        0      0 0.0.0.0:36466           0.0.0.0:*                           2016/rsyslogd
udp        0      0 0.0.0.0:53112           0.0.0.0:*                           3339/rsyslogd
udp        0      0 0.0.0.0:53122           0.0.0.0:*                           5607/rsyslogd
udp        0      0 0.0.0.0:36741           0.0.0.0:*                           3227/rsyslogd
udp        0      0 0.0.0.0:53640           0.0.0.0:*                           2872/rsyslogd
udp        0      0 0.0.0.0:53894           0.0.0.0:*                           2665/rsyslogd
udp        0      0 0.0.0.0:46166           0.0.0.0:*                           3140/rsyslogd
udp        0      0 0.0.0.0:38519           0.0.0.0:*                           1105/rsyslogd
udp6       0      0 :::123                  :::*                                3194/ntpd
```

Issue: some applications open up the port without specifying the address

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Security Analysis

- Security Vulnerabilities – Port scan reveals lots of processes using 0.0.0.0 address
- Redist db a single point of failure
  - 32 bit system that manages configdb, state db, routing and SAI
- Limited ACL Support
  - Limited support for Filtering rules in Hardware

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